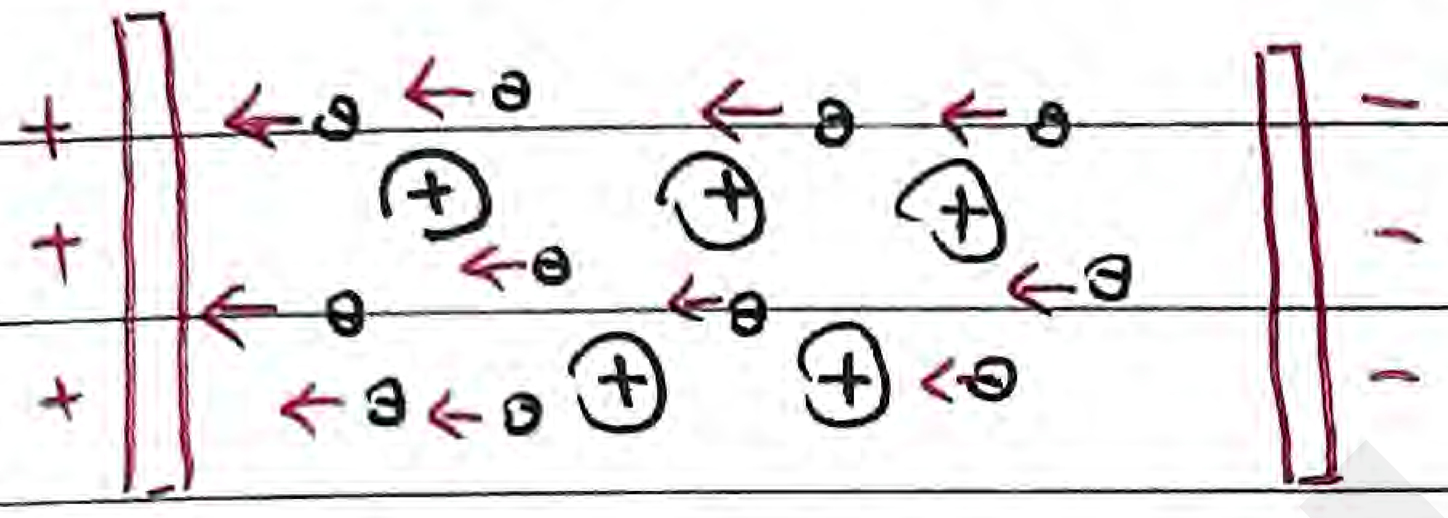
Electric current

when do charges move?

- when there's a force exerted on the charge
- In other words, when there's an electric field

Let's consider a lattice structure



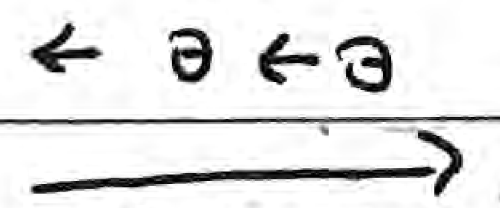
Let's add a uniform electric field

Adding a uniform electric field means, probabistically, most electrons flow in the same direction

- Only electrons move because of lattice structure (more energetically favorable)

* Electrons do not move linearly due to electron-electron repulsion and electron-nucleus attraction. It's more zig-zag than linear.

The direction of electric current is defined as that opposite to the motion of electrons



I = electric current

Electrons also move really slowly ($10^{-2} ms^{-1}$ magnitude)

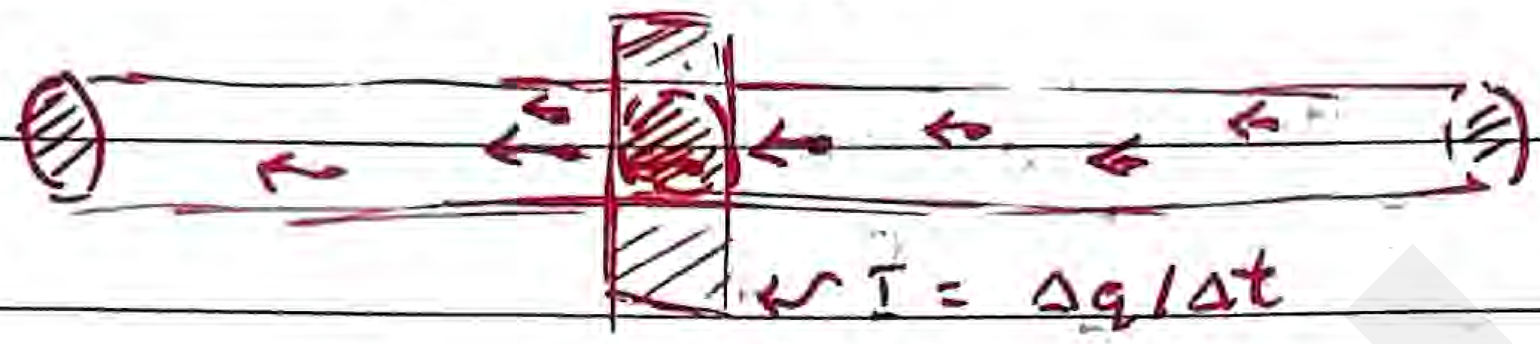
- However, electricity is instant due to fields, which instantly exert force on e^-

Date. NO.

Electric current

Defined as change in charges per reference area in a given time unit (second)

$$I = \frac{\Delta q}{\Delta t}$$

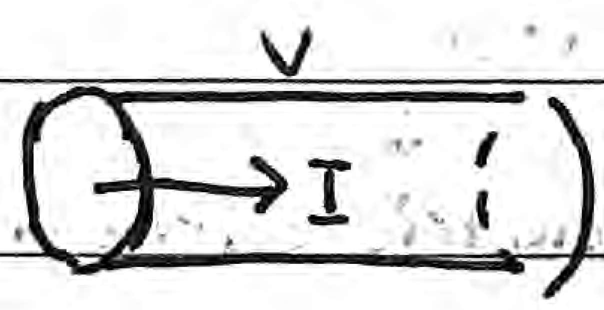


where units are $C \cdot s^{-1} = A = \text{Ampere}$

Electric resistance

While friction certainly does not exist in a subatomic level, there's repulsive and resistive forces against electrons.

This is known as resistance, defined as the potential difference over the current in a given conductor

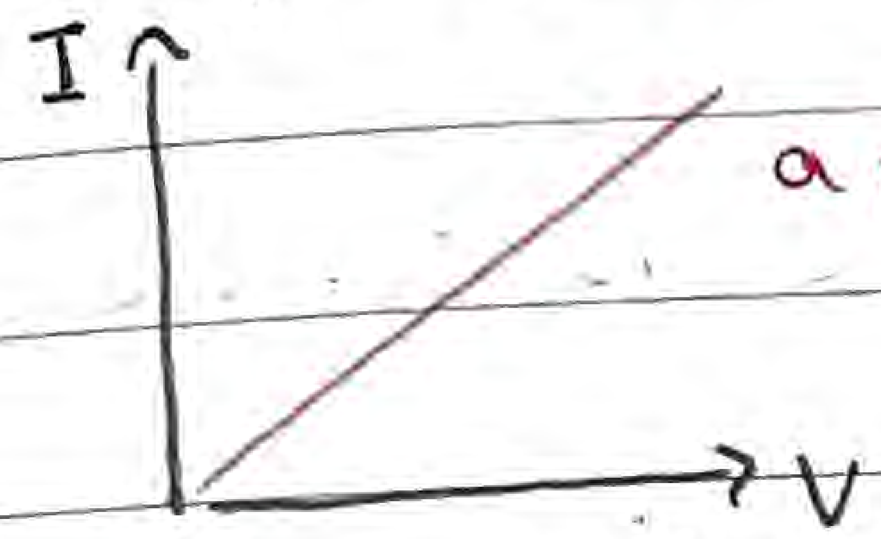


$$R = \frac{V}{I} \quad [V \cdot A^{-1} = \Omega = \text{Ohm}]$$

Electric resistance

When you measure current per different values of voltage, you can calculate the resistance of the material upon a change in voltage (applied)

- Graphing them out:

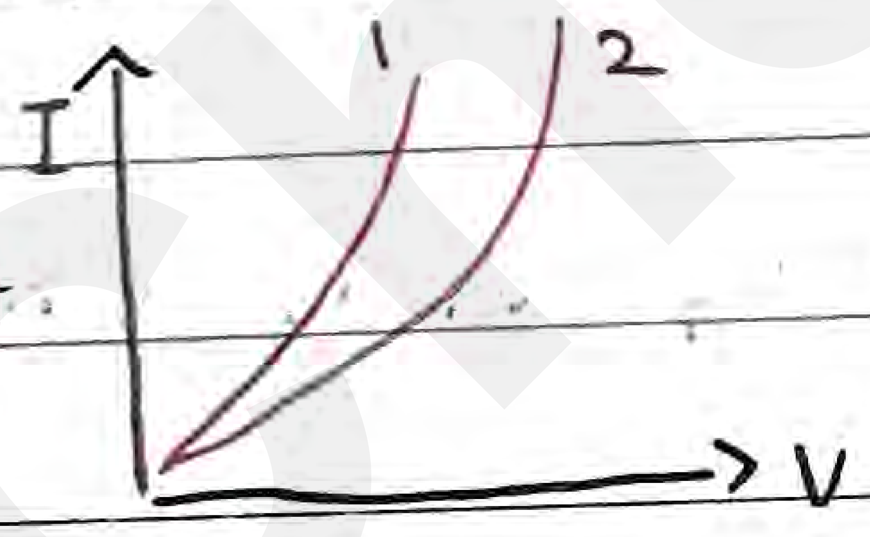


a perfectly linear relation

constant resistance →

This is an ohmic material (e.g. resistors)

However... some materials do not have a perfectly linear relation

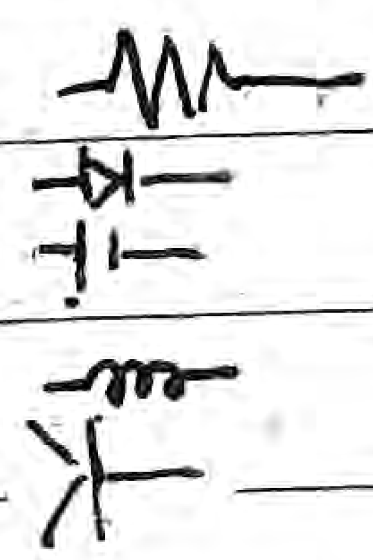


non-constant resistance

This is a non-ohmic material
- All natural materials are non-ohmic

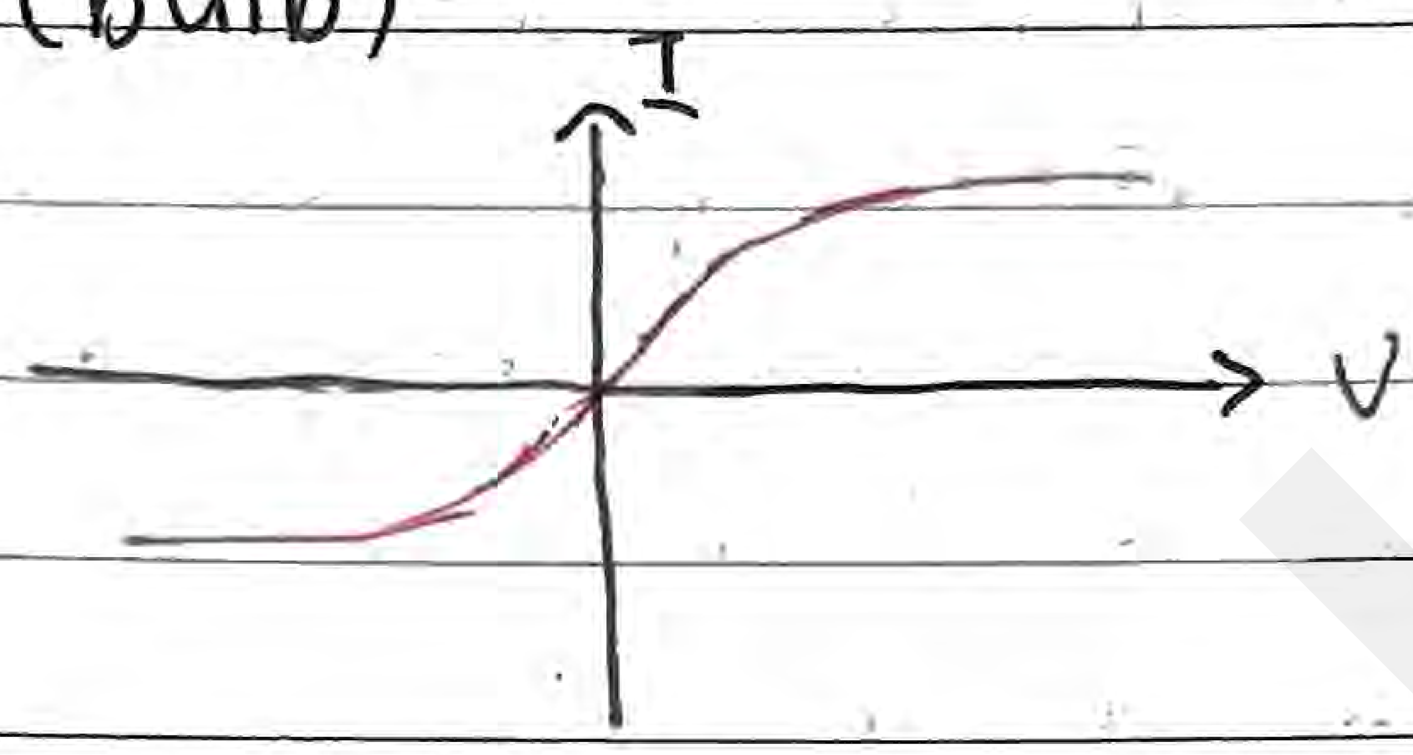
In circuits, everything is composed of these five components:

1. Resistor
2. Diode
3. Capacitor
4. Coil / Inductors
5. Transistors

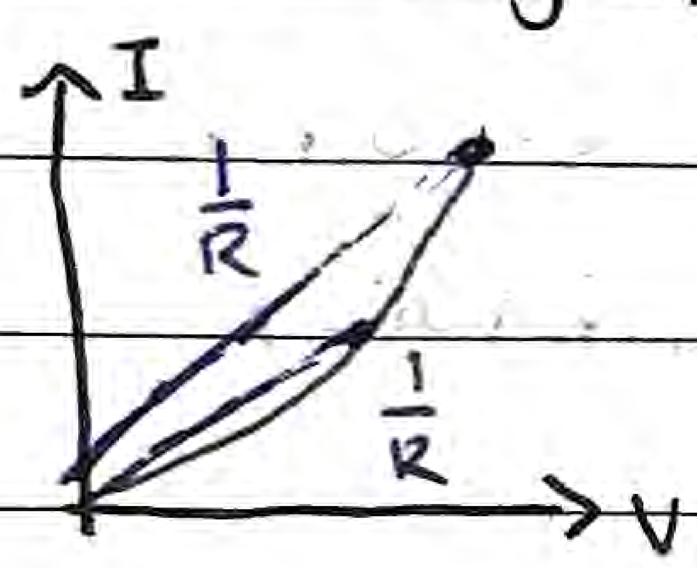


I-V graph

Filament (bulb)



In a I-V graph, resistance is NOT the slope



$R = \frac{V}{I} \neq \frac{\Delta V}{\Delta I}$

it's the reciprocal of the average rate of change from $V = 0$ (not the instantaneous!)

Resistance is affected by various factors

The following formula illustrates this relationship
↑ for a conductor

$$R = \rho \cdot \frac{L}{A}$$

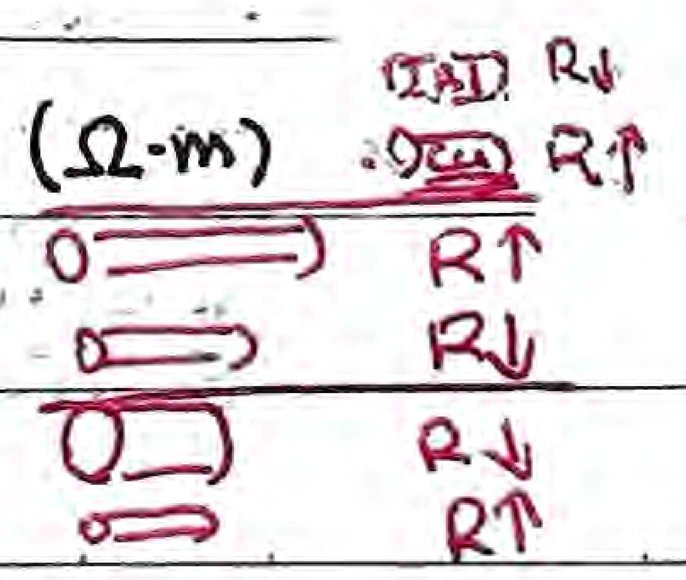
where ...

R - Resistance of object

depends on temperature → $\rho(T)$ - Resistivity of material ($\Omega \cdot m$)

L - Length of object

A - Cross-sectional area



Electrical Conductivity

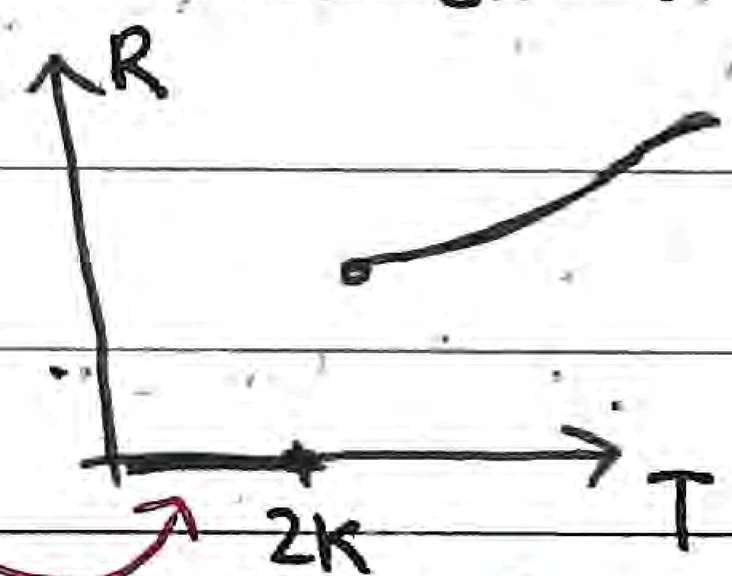
$$\text{Electrical Conductivity} = \frac{1}{R}$$

more resistive, less conductive
less resistive, more conductive

Temperature and resistance

As temperature increases, resistance increases.
Lower temperature, lower resistance

but at some point, resistance jumps to 0



resistance reaches 0Ω
This is a superconductor (perfect for efficiency)

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Electricity

Electric circuit - To transfer energy from source to consumer

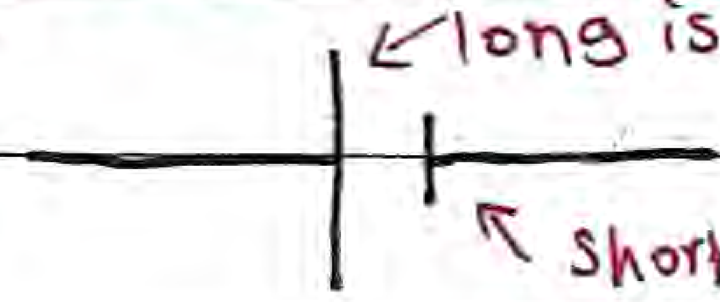
Source of energy —transfer of energy→ Conservation of energy

but you also need to close the loop!

In order to create a potential difference, you need a "pump" or a battery

Electric Circuit Symbols

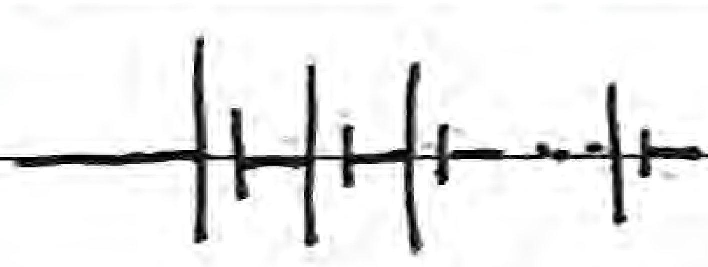
Source of energy



← long is + ↖ short is -

Cell (converts chemical energy to electrical)

(solar panel is not included here)



Battery (multiple cells)

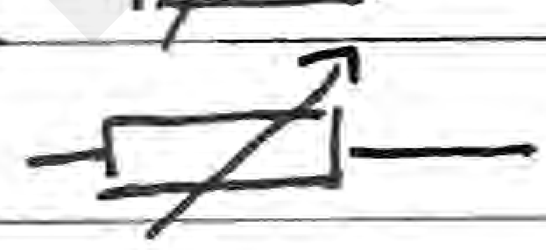
cons of energy

converts electric energy into thermal energy

Resistor

Lamp

thermistor (resistor, resistance changes to temperature)



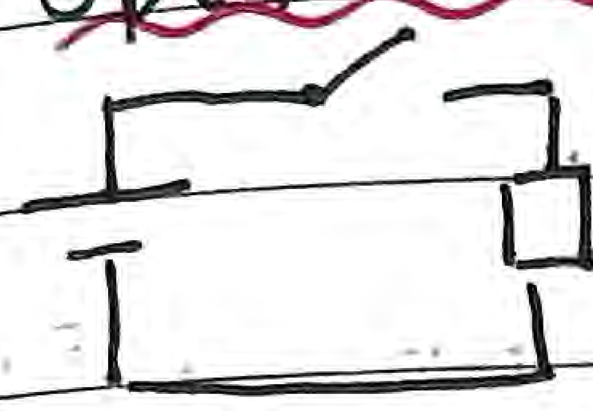
variable resistor (resistor, manually (mechanically) changes resistance

Switch (opens/closes circuit for current)

Wire

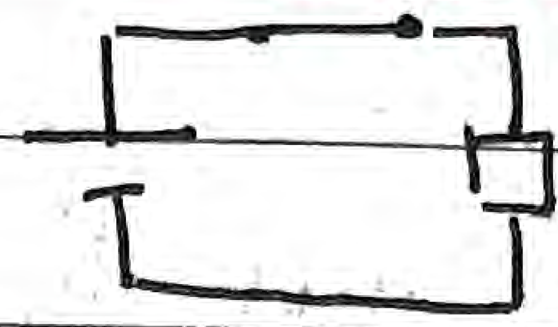
Basic Electric circuit

Open circuit

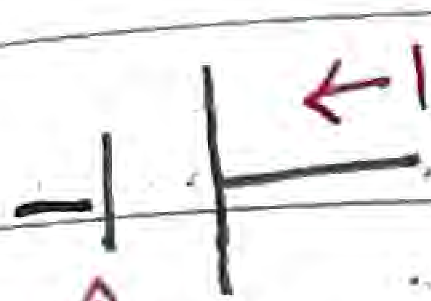


$I = 0$

Closed circuit



opposes flow of electron → Current is positive to negative end



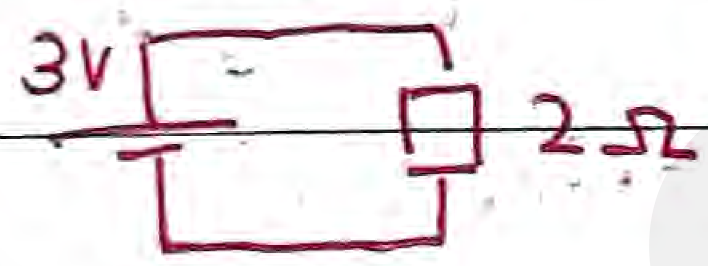
← long line is POSITIVE end

short line is NEGATIVE end

Ohm's law:

$$R = \frac{V}{I} = \frac{\text{Potential Difference}}{\text{current}}$$

E.g.



Then current $= \frac{3}{2} A = 1.5A$

However, circuits are not that simple...

Resistors in series

terminals do not meet/ not the same →

Equivalence Resistance $= \sum R_i = R_1 + R_2 + R_3 + \cdots$

Total Voltage $= \sum V_i = V_1 + V_2 + V_3 + \cdots$

- The current through each resistor is the same
- As such, according to Ohm's law, the Potential Difference on each resistor is not the same

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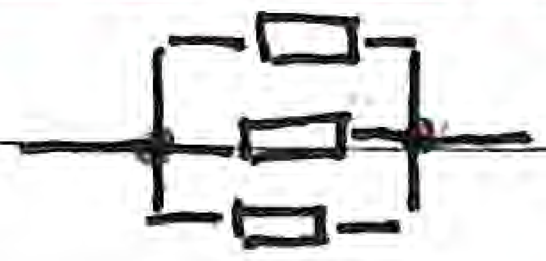
Resistors in parallel



All three terminals of the resistors meet / are of the same point

- The potential difference on each resistor is the same
- The Equivalent Resistance is calculated as follows:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n} = \sum \frac{1}{R_i}$$

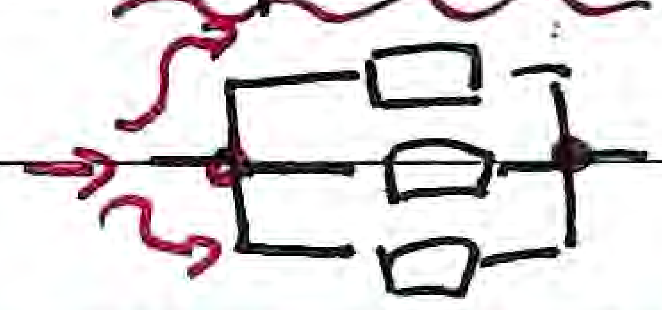
- According to Ohm's law, the current on each resistor is not the same: $I = I_1 + I_2 + I_3 \cdots = \sum I_i$

You can think of it like this

In series

- Same "wire", same current
- Different terminal's, different voltages

In parallel



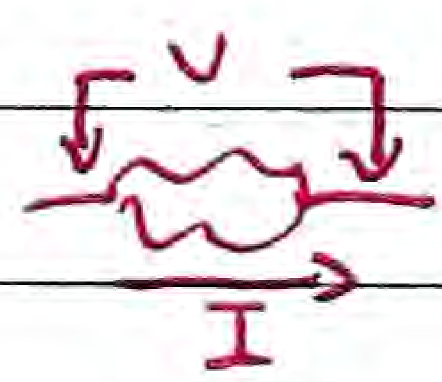
- Different "wire", different currents
- Same terminal, same voltage

Electric Power

- You can calculate the energy passing through any component in a given amount of time
  - ⇒ Depending on whether it is source or consumer of energy, it will be power released or absorbed respectively

total power = $\sum$ of P_i (for both series, parallel)

$\boxed{P = I^2 \cdot R} = V \cdot I$ → $V = I \cdot R$ → $\boxed{P = \frac{V^2}{R}}$



where...

P – Electric power of component

R – Resistance of component

I – Current passing through

V – Potential difference of terminals

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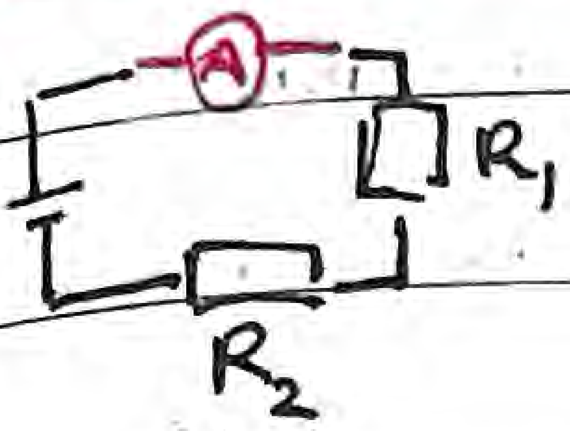
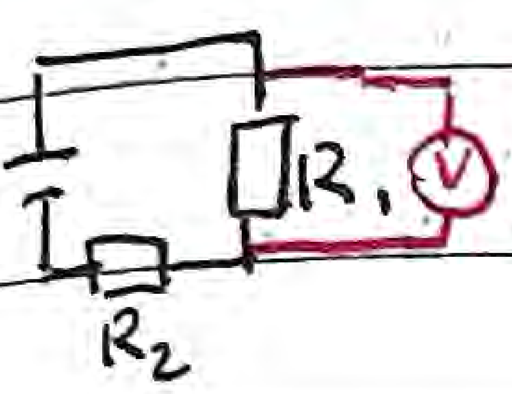
Ideal Ammeter - An instrument measuring the current in a series such that the resistance is negligible



↳ to allow for all current to pass through ammeter

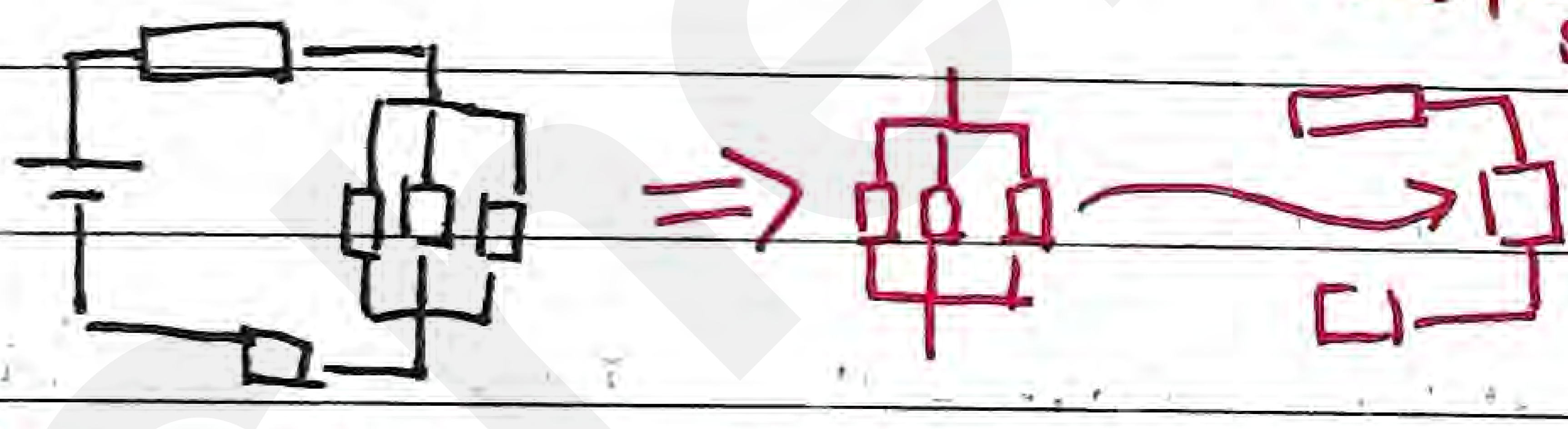
Ideal Voltmeter - An instrument measuring the potential difference in parallel such that the resistance is infinitely large



↳ to prevent current from escaping to voltmeter

Complex circuits often consist of in series and in parallel components

↳ You can group these different components to do mini calculations (go from innermost circuit, the one you're 100% certain with, of the same type)

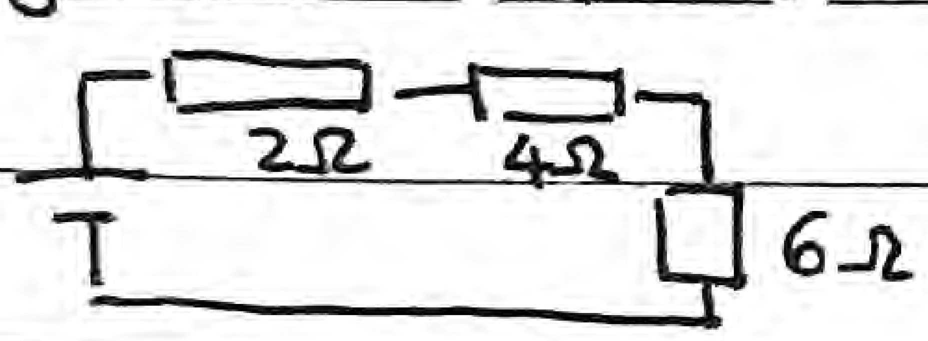


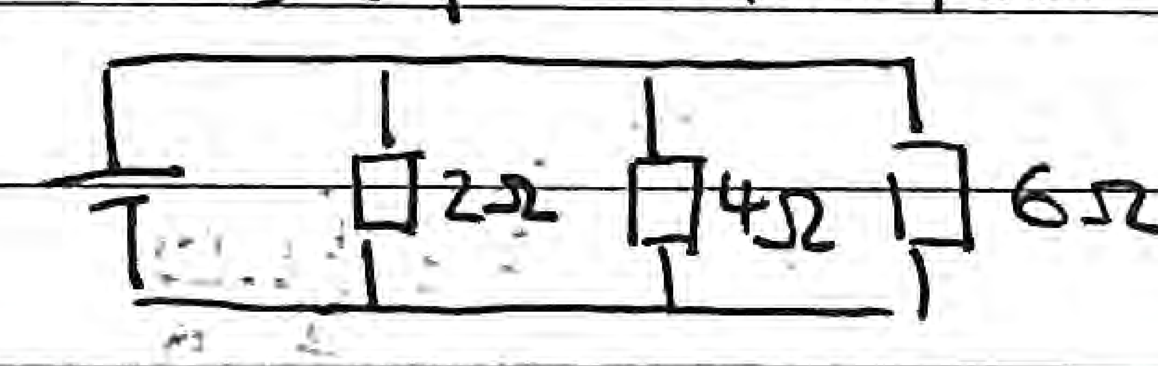
to then calculate R_{eq} for overall circuit

· After you find current of entire circuit, you go inwards one-by-one to

Analyzing parallel and series circuits

E.g. For 24V circuit with resistors of 2Ω, 4Ω, 6Ω





R_{eq} lower for parallel, greater current, greater power (useful for high power)

R_{eq} higher for series, lower current, lower power (useful for low power)

Date. NO.

Types of resistors

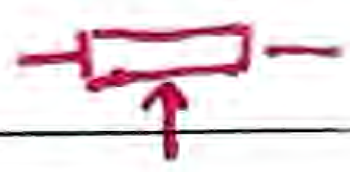
> Thermistor – Temperature dependent resistance

- As temperature increases, resistance decreases

> Light dependent resistor (LDR) – Intensity of light dependent resistance

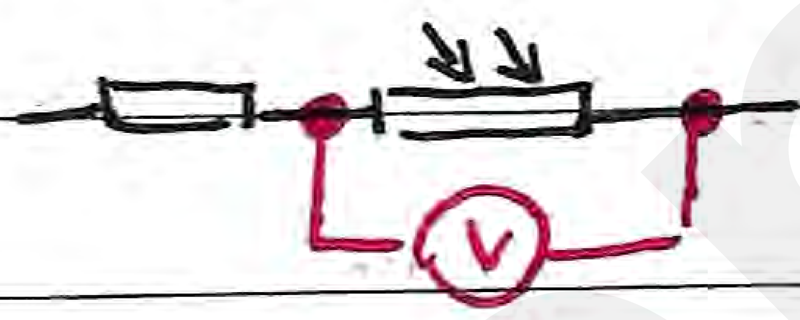
- As intensity of light increases, resistance decreases

> Potentiometer – Length dependent resistance



- As length of resistor increases, resistance increases

E.g. Given constant current, what happens to measured voltage as light intensity increases



$I_L \uparrow$ thus $R \downarrow$

Given $V = I \cdot R$

$V \propto R \rightarrow R\downarrow$ so $V\downarrow$

Voltage, current, resistance

- As we know, voltage stays constant in a circuit connected in parallel, current stays constant in a circuit connected in series.
- As such, we can also get the following formula...

$V = IR$

$\hookrightarrow \dfrac{V_1}{V_2} = \dfrac{I_1}{I_2} \cdot \dfrac{R_1}{R_2}$

For constant voltage (parallel)

$$\frac{I_2}{I_1} = \frac{R_1}{R_2}$$

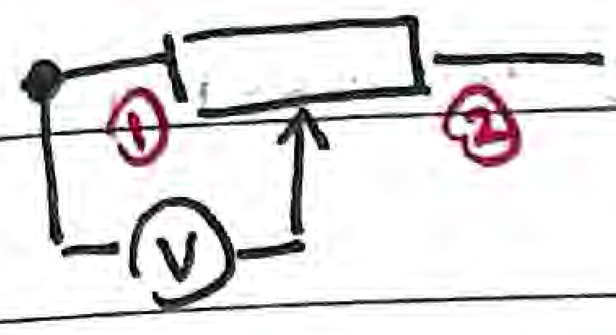
For constant current (series)

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

Can be useful when one resistance is equivalence resistance

Potentiometer

E.g.

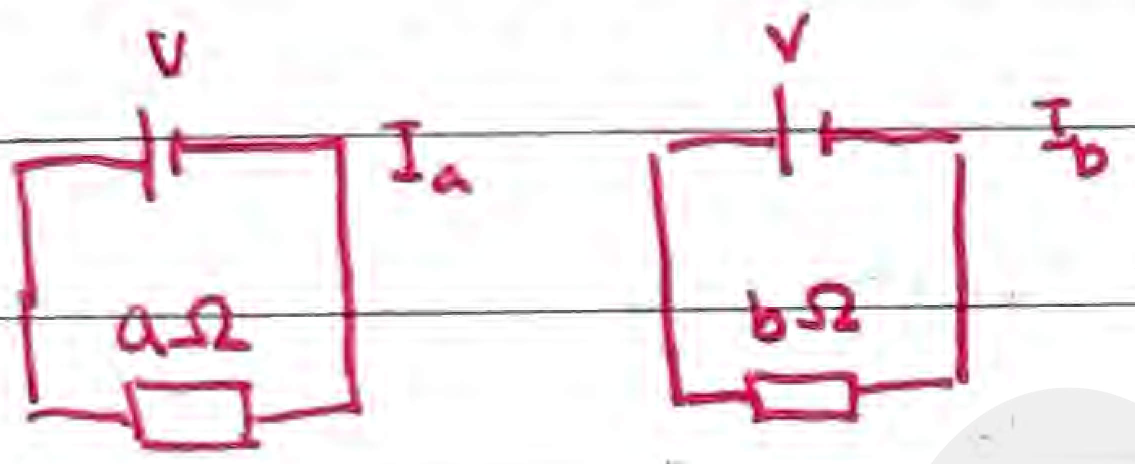


the labelled resistance is the maximum resistance

- At ①, voltmeter does not measure a potential difference, V = 0
- At ②, voltmeter measures maximum potential difference

Voltage, current, resistance

- When resistance changes in a circuit, so will the current (assuming voltage is constant due to cell)



if b > a, then $I_b < I_a$

- So, when a problem requires a change in resistance, make sure to recalculate the current
  ↳ To find voltage, power, etc.

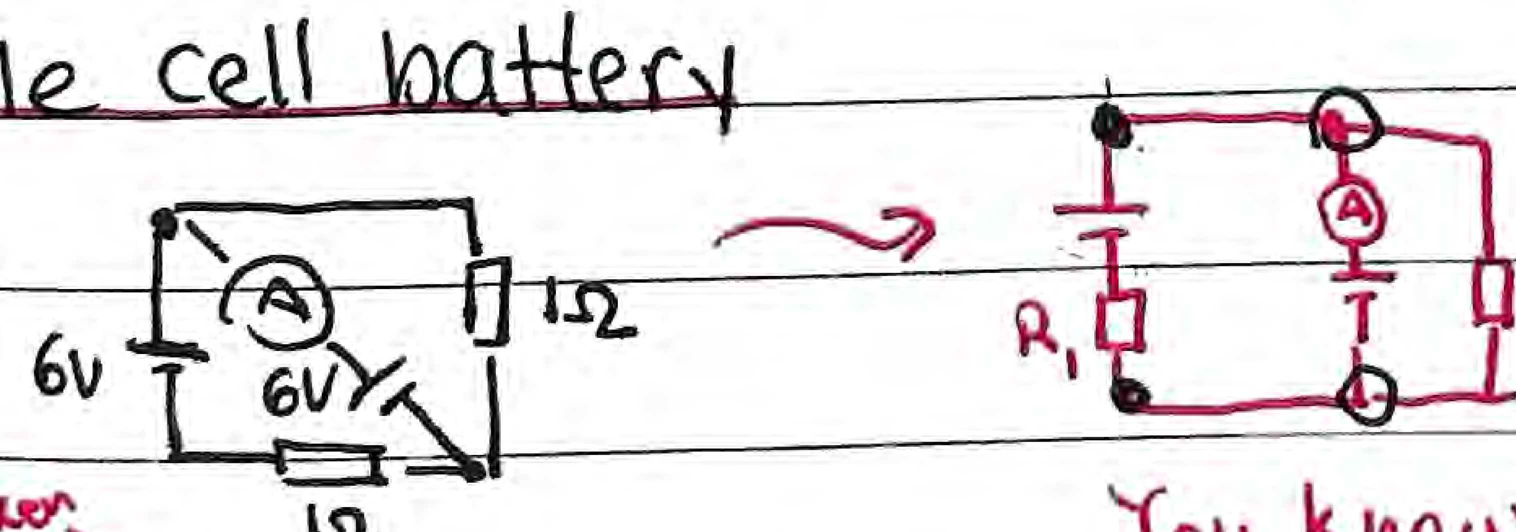
Power consumed = Power delivered

ΣP_i of individual circuits = Power of system

$= I \cdot V$

Multiple cell battery

E.g.



two terminals are connected in parallel, V is same

You know V of ⚬⚬ is 6V, but not : (since there's resistance)

R_1 must have 0 A passing through ← thus also 6V ←

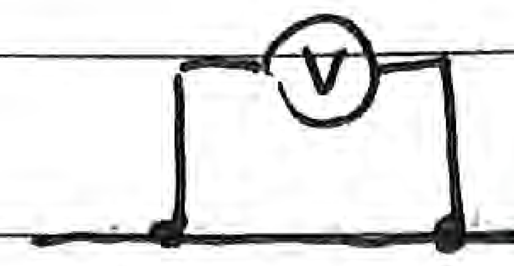
(or else would energy decrease (taken out))

Date. ______ NO. ______

Complicated electric circuits

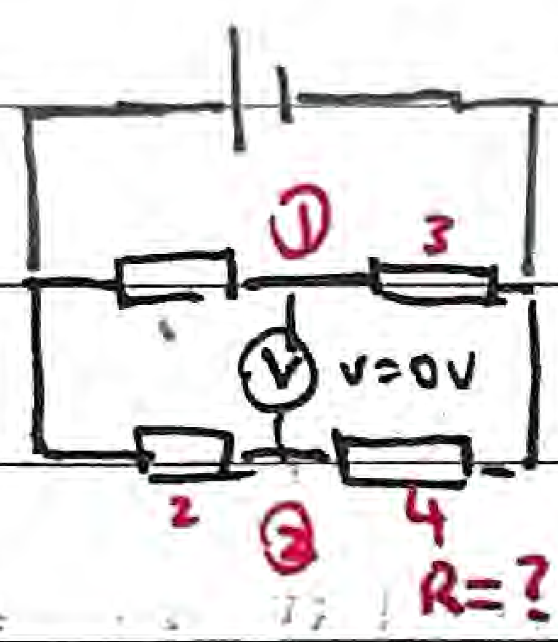
· Remember an ideal voltmeter has ∞ resistance

· And, when...

$V = 0\ V$ 

is the same as no resistor in between

then... E.g.



→ $V=0$ so energy from cell to 1 is same as to 2

thus $V_1 = V_2$, $V_3 = V_4$

but also, voltmeter allows no current to pass through, so $I_3 = I_1$ (no splitting), $I_4 = I_2$

thus $\frac{V_1}{V_2} = \frac{V_3}{V_4} \Rightarrow \frac{I_1 R_1}{I_2 R_2} = \frac{I_3 R_3}{I_4 R_4}$

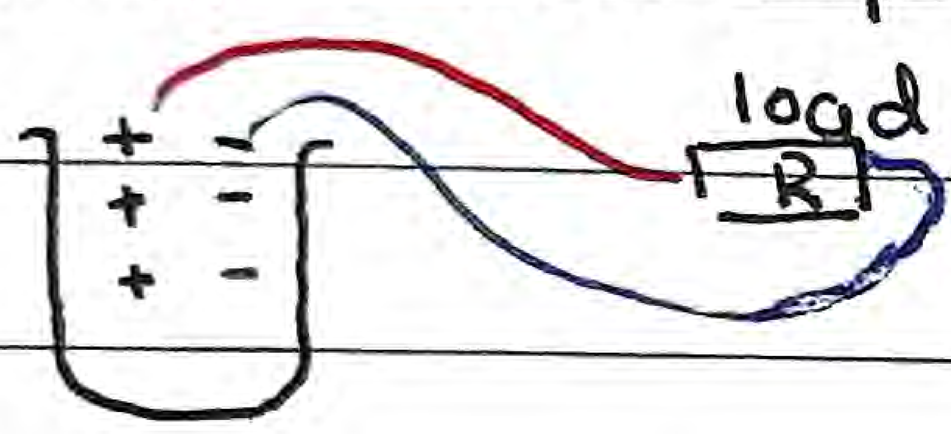
Electromotive Force (Emf, ε)

→ Not force, it's energy

The work done per unit charge in moving a charge from one terminal of the battery to the other.
↳ the units are volts, but it's not the same as electrical potential difference.

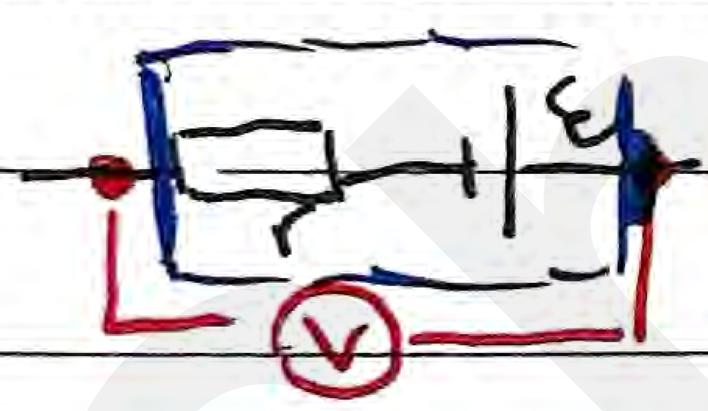
Internal Resistance

Recall a battery...



since charges must travel through a medium (non-vacuum)

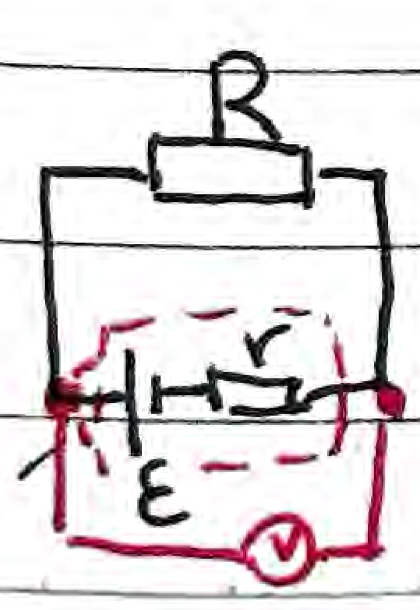
Thus we can think of it like this...



Real Battery

Ideal Battery

ε is the voltage of an ideal battery and the voltage of a real battery when current = 0 (open circuit)
↳ Thus actual voltage of a real battery in a closed circuit will never reach ε (advertized voltage)



current = $\frac{\varepsilon}{R+r}$ $\varepsilon = I(R+r)$

Voltage = $\varepsilon - Ir$

Energy taken out

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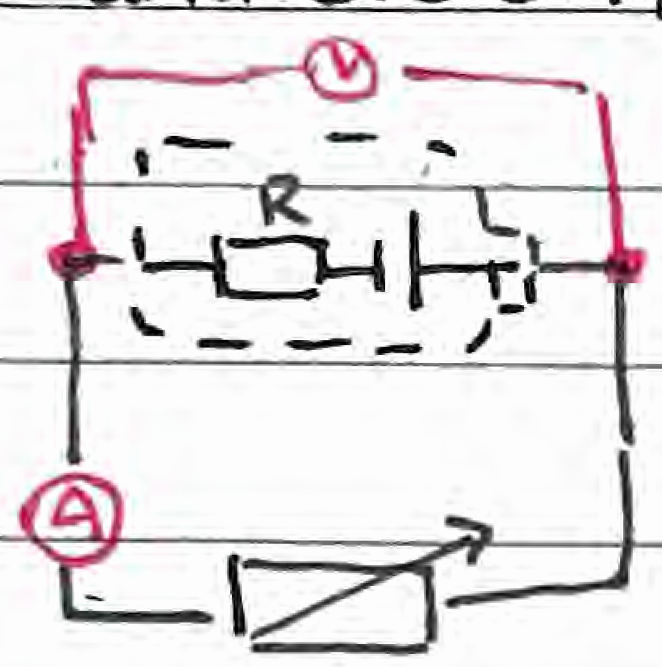
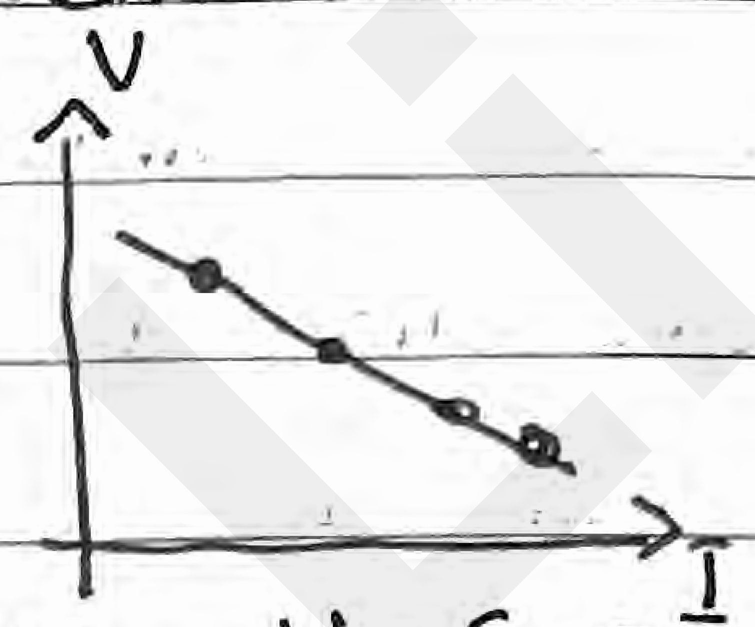
Measuring internal resistance

- You can measure internal resistance by changing external resistance and measuring current and electrical potential difference



I vs V

I	V
-	-
-	-
-	-



since $V = \varepsilon - Ir$

so Emf is y-intercept, internal resistance is slope